

Object Segmentation to prevent contextual shortcut learning

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Abstract

Deep learning models can rely on background information as a shortcut instead of learning robust object-specific features. This project investigates whether automatic foreground segmentation can reduce such shortcut learning in a 10-class iNaturalist-12K image classification task. Foreground masks were generated with SAM and CLIP, and CNN classifiers were trained with different probabilities of replacing original images by segmented versions. Partial segmentation reduced the train-validation gap during model selection, suggesting a regularizing effect, but did not consistently improve final test performance. Full segmentation substantially degraded performance, likely due to imperfect masks and loss of useful visual context. A pretrained ResNet18 clearly outperformed the custom CNN, showing the benefit of transfer learning on this relatively small dataset.

1 Introduction

Deep learning models often perform well on benchmarks but may rely on *shortcuts*, i.e. decision rules that fail under challenging or out-of-distribution conditions [2]. This is problematic in settings such as autonomous driving, where models must generalize to unseen environments [1]. At the same time, prior work has shown that image backgrounds can contain highly predictive information and may substantially contribute to classification accuracy [11].

This project investigates whether automatic foreground segmentation can improve robustness and reduce shortcut learning in biological image classification using a 10-class subset of iNaturalist-12K [6]. A segmentation pipeline based on Segment Anything (SAM) [5, 8] and CLIP [7] was developed to automatically extract the foreground object without requiring manual annotations. The resulting segmented masks were used to crop images to the region of interest and to train CNN-based classification models. The performance of models trained with this segmentation-based regularization approach was then compared against equivalent models trained on the original, non-segmented images.

2 Related work

Shortcut learning is a limitation of deep neural networks [2]. In ecological datasets, background information can become a strong shortcut because images from the same location often share similar environmental context, particularly in datasets collected from fixed animal camera traps [1]. Geirhos et al. argue that shortcut learning should not only be studied through in-distribution accuracy, but rather by manipulating the evidence available to the model, such as background, texture, or color, and observing how predictions change [2]. Several studies demonstrate that image backgrounds alone can contain sufficient information for classification [11, 12]. Xiao et al. showed that models trained on ImageNet often strongly

depend on contextual background information, while Rosenfeld et al. demonstrated that perturbations can alter predictions [10, 11]. Geirhos et al. further found that CNNs frequently rely more on texture cues than on object shape, which may contribute to such behavior and provides additional motivation for segmentation-based approaches that emphasize object structure over contextual texture information [3].

These findings motivate approaches that explicitly separate foreground objects from their surrounding context. However, many supervised segmentation methods, including segmentation-based variants of YOLO, require manually annotated masks and labels for training, which is highly labor-intensive and not given for the iNaturalist dataset [9]. Segment Anything (SAM) [5, 8] provides automatic segmentation proposals without requiring manual annotation, while CLIP can score image regions semantically using text-image similarity [7]. Such a pipeline is used in this project and is conceptually related to copy-paste augmentation methods [4], where segmented foreground objects are inserted into alternative backgrounds to improve robustness and reduce contextual bias.

3 Experiments

Dataset. The dataset used for this project is a 10-class subset of the iNaturalist-12k dataset available on Kaggle (link in appendix), provided by Aryan Pandey. It contains 12,000 images across biological categories such as Amphibia, Mammalia, and Plantae. The classes are balanced with 1,200 available images for each of the classes. The dataset was split into 8,000 training images, 2,000 validation images, and 2000 test images, stratified by class.

Model architecture. Figure 1 shows the custom CNN used in this work. It consists of four convolutional blocks, each applying two 3×3 convolutions with batch normalization and ReLU, followed by 2×2 max pooling. Global average pooling and a small fully connected classifier produce the final 10-class logits. As a transfer-learning baseline, a pretrained ResNet18 with ImageNet weights was used by replacing its final layer with a new 10-class classifier.

Training Details. Models were trained with cross-entropy loss, AdamW, batch size 32, image size 224×224 , random resized crops, horizontal flips, dropout 0.3, weight decay, and gradient clipping at 1.0. Early stopping was used during model selection. Final training used the selected epoch counts without early stopping: 122 epochs for the custom CNN and 30 for ResNet18. For the custom CNN, the final setting used learning rate 0.001 and weight decay $3e-4$. ResNet18 used learning rate 0.001 and weight decay $1e-4$. Each architecture was finally trained with segmentation probabilities $p \in \{0.0, 0.25, 1.0\}$, where p denotes the probability of replacing an original training image by its segmented version.

Results. The segmentation masks were manually evaluated on 200 images, with 20 samples per class. A mask was counted as good if it

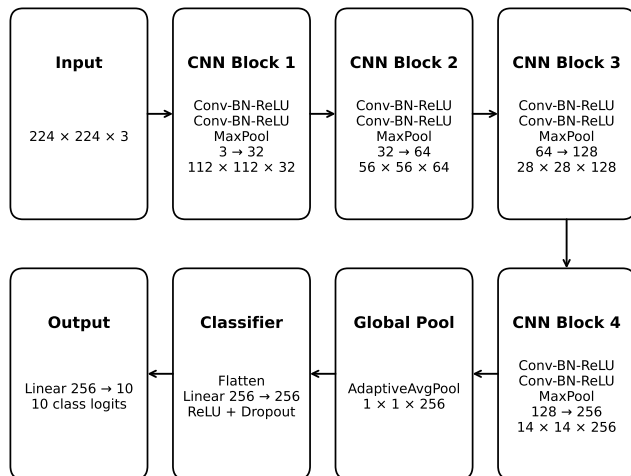


Figure 1: Architecture of custom model

captured the main foreground object without large missing regions or excessive background. Mask quality varied strongly between classes: masks were weak for *Animalia*, but reliable for *Aves* (see results in table 2 of appendix).

Table 1: Final test results for ResNet18 and the custom CNN under different segmentation probabilities.

Architecture	Seg. prob.	AUC	Recall	Precision
ResNet18	0.00	0.963	0.740	0.741
ResNet18	0.25	0.965	0.746	0.750
ResNet18	1.00	0.944	0.674	0.680
Custom CNN	0.00	0.887	0.544	0.554
Custom CNN	0.25	0.882	0.535	0.538
Custom CNN	1.00	0.697	0.241	0.278

During model selection, partial segmentation behaved like a regularizer. For the custom CNN, the non-segmented baseline reached train accuracy 0.700 and validation accuracy 0.551 at its best epoch, while training with partial segmentation reduced the training accuracy to about 0.607 for $p = 0.2$ and 0.606 for $p = 0.5$, with similar but slightly lower validation accuracy around 0.546 and 0.541. This indicates a smaller train-validation gap, but no clear validation-performance gain. In contrast, full segmentation ($p = 1.0$) performed poorly, reaching only 0.255 validation accuracy.

The final test results in Table 1 show the same overall pattern. Partial segmentation ($p = 0.25$) produced almost identical performance to training on original images. For ResNet18, the differences between $p = 0.0$ and $p = 0.25$ are very small and should not be interpreted as a clear improvement. For the custom CNN, partial segmentation slightly worsened AUC, recall, and precision. Full segmentation clearly degraded both models, especially the custom CNN. Overall, ResNet18 performed substantially better than the custom CNN.

Figure 2 shows the class-wise metric differences between $p = 0.25$ and $p = 0.0$. No consistent class-level trend is visible. Even

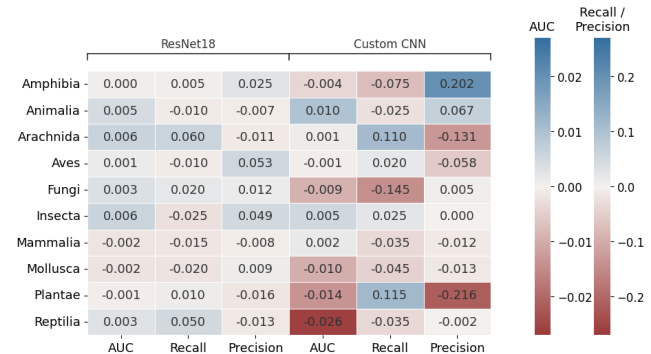


Figure 2: Comparison of models trained with partial segmentation ($p = 0.25$) against models trained on original images. Positive values indicate an improvement from segmentation; negative values indicate worse performance.

Aves, which had the best mask quality, did not clearly benefit from segmentation. Some classes improved in one metric but worsened in another, such as *Plantae* and *Arachnida*. This suggests that partial segmentation acted mainly as a weak regularizer rather than as a systematic performance improvement.

4 Discussion

This project investigated whether automatic foreground segmentation can reduce background shortcuts in biological image classification. The results suggest that partial segmentation can act as a mild regularizer, since it reduced overfitting during training, but it did not consistently improve final test performance. Full segmentation performed poorly, probably because the automatically generated masks were imperfect and removed useful visual context in addition to background information.

A key limitation is the quality of the segmentation pipeline. Since masks were generated automatically with SAM and selected using CLIP, errors in mask selection directly affected the training data. Even classes with visually good masks, such as *Aves*, did not clearly benefit, suggesting that background removal alone is not sufficient in this setup. The study was also limited by computational and temporal resources, a small number of final runs, and a relatively small dataset for training CNNs from scratch. This is also reflected by the strong performance gap between the custom CNN and pretrained ResNet18. Future work should repeat the experiment with cleaner segmentation labels, stronger validation across seeds, and controlled background replacement, for example by inserting segmented objects into unusual contexts to directly test shortcut reliance.

5 References

References

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Table 2: Manual quality assessment of segmentation masks by class (n=200). Values indicate the percentage of masks rated as good.

Class	Amph.	Anim.	Arach.	Aves	Fungi
Good masks (%)	50	29	43	94	79
Class	Insecta	Mamm.	Moll.	Plant.	Rept.
Good masks (%)	74	81	74	76	67

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A Dataset

<https://www.kaggle.com/datasets/aryanpandey1109/inaturalist12k>

B Results

C Prompts for SAM

D Declaration of generative AI usage

I hereby declare that I have used an LLM (ChatGPT, model GPT-5.3 and GPT-5.5) to assist me with some small passages of the code, debugging, improve some readability, to get new ideas, to assist me with training on the HPC server, to visualize the model architecture, and to find better vocabulary and restructuring some sentences while writing the paper. This is in accordance with the "Guidelines on the Use of Generative AI for Teaching and Learning", Version: 1.0, Date: 2026-02-16.